

Paste Key:

$\Sigma, \text{ä}, \text{b}, \text{c}, \in, \equiv, \oplus$

HW3: Block Cyphers

CYB-402

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1. Generate the multiplication table for the extension field $GF(2^3)$ for the case that the irreducible polynomial is $P(x) = x^3 + x + 1$. The multiplication table is in this case a 8×8 table.

$GF(2^3)$, $P(x) = x^3 + x + 1$.

$ax^2 + bx + c, \dots, a, b, c \in \{0,1\}$

Vector representation: (a,b,c)

Polynomial	Vector
0	(0,0,0)
1	(0,0,1)
x	(0,1,0)
x + 1	(0,1,1)
x ²	(1,0,0)
x ² + 1	(1,0,1)
x ² + x	(1,1,0)
x ² + x + 1	(1,1,1)

$$A(x) * B(x) = (x^2 + x + 1)(x^2 + 1)$$

$$= x^4 + x^3 + x^2 + x^2 + x + 1$$

$$= x^4 + x^3 + (1 + 1)x^2 + x + 1 \pmod{2}$$

$$= x^4 + x^3 + x + 1$$

$$P(x) = x^3 + x + 1 = x^3 = x - 1, \text{ using reduction rule}$$

$$= x^3 \equiv x + 1$$

From this we can get the rest,

$$x^4 = x * x^3 = x(x + 1) = x^2 + x$$

$$x^5 = x * x^4 = x(x^2 + x) = x^3 + x^2 = x^2 + x + 1$$

$$x^6 = x * x^5 = x(x^2 + x + 1) = x^3 + x^2 + x = x + 1 + x^2 + x \pmod{2}$$

$$= x^2 + 1$$

$$x^7 = x * x^6 = x(x^2 + 1) = x^3 + x = x + 1 + x \pmod{2} = 1$$

x generates all nonzero elements

	0	1	x	x + 1	x ²	x ² + 1	x ² + x	x ² + x + 1
0	0	0	0	0	0	0	0	0
1	0	1	x	x + 1	x ²	x ² + 1	x ² + x	x ² + x + 1
x	0	x	x ²	x ² + x	x + 1	1	x ² + x + 1	x ² + 1
x + 1	0	x + 1	x ² + x	x ² + 1	x ² + x + 1	x ²	1	x
x ²	0	x ²	x + 1	x ² + 1	x ² + x	x	x ² + 1	1
x ² + 1	0	x ² + 1	1	x ²	x	x ² + x + 1	x + 1	x ² + x
x ² + x	0	x ² + x	x ² + x + 1	1	x ² + 1	x + 1	x	x ²
x ² + x + 1	0	x ² + x + 1	x ² + 1	x	1	x ² + x	x ²	x + 1

$$x^3 + 1 = x + 1 + 1 \pmod{2} = x$$

$$x(x^2 + 1) = x^3 + 1 = x$$

$$(x+1)(x^2 + x + 1) = (x^3 + x^2 + x) + (x^2 + x + 1) = x^3 + (1 + 1)x^2 + (1+1)x + 1 \pmod{2}$$

$$= x^3 + 1 = x$$

2. Compute A(x) + B(x) mod P(x) in GF (2⁴) using the irreducible polynomial P(x) = x⁴ + x + 1. What is the influence of the choice of the reduction polynomial on the computation?

The choice of P(x) doesn't affect addition. So, there is no influence.

2.1. A(x) = x² + 1, B(x) = x³ + x² + 1

$$(x^2 + 1) + (x^3 + x^2 + 1) = x^3 + (1+1)x^2 + (1+1) \pmod{2} = x^2$$

2.2. A(x) = x² + 1, B(x) = x + 1

$$(x^2 + 1) + (x + 1) = x^2 + x + (1 + 1) \pmod{2} = x^2 + x$$

3. Compute A(x) · B(x) mod P(x) in GF (2⁴) using the irreducible polynomial P(x) = x⁴ + x + 1. What is the influence of the choice of the reduction polynomial on the computation?

The choice of P(x) matters, different irreducible polynomials will give different reduction identities and different multiplication tables.

Using P(x) = x⁴ + x + 1; we use reduction rule to get

$$x^4 \equiv x + 1$$

$$x^5 = x * x^4 = x(x+1) = x^2 + x$$

3.1. A(x) = x² + 1, B(x) = x³ + x² + 1

$$(x^2 + 1)(x^3 + x^2 + 1) = x^5 + x^4 + x^3 + (1+1)x^2 + 1 \pmod{2} = x^5 + x^4 + x^3 + 1$$

$$x^5 + x^4 + x^3 + 1 \equiv (x^2 + x) + (x + 1) + x^3 + 1 = x^3 + x^2 + (1+1)x + (1+1) \pmod{2} = x^3 + x^2$$

$$\mathbf{3.2. A(x) = x^2 + 1, B(x) = x + 1}$$

$$(x^2 + 1)(x + 1) = x^3 + x^2 + x + 1$$

Degree < 4, no further reduction needed.

4. Compute in GF(2⁸): (x⁴ + x + 1)/(x⁷ + x⁶ + x³ + x²), where the irreducible polynomial is the one used by AES, P(x) = x⁸ + x⁴ + x³ + x + 1. Note that Table 4.2 in the textbook contains a list of all multiplicative inverses for this field.

$$P(x) = x^8 + x^4 + x^3 + x + 1$$

$$\text{Need } A(x) * B(x)^{-1} \pmod{P(x)}$$

$$A(x) = x^4 + x + 1 = (00010011) = 13_{16}$$

$$B(x) = x^7 + x^6 + x^3 + x^2 = (11001100) = CC$$

$$\text{Need } B(x)^{-1}; \text{ from table } B^{-1}(x) = 1B = (00011011) = x^4 + x^3 + x + 1$$

Check:

$$B(x) * B(x)^{-1} = 1$$

$$B(x) * B(x)^{-1} = (x^7 + x^6 + x^3 + x^2) (x^4 + x^3 + x + 1) =$$

$$x^{11} + x^{10} + x^8 + x^7 + x^{10} + x^9 + x^7 + x^6 + x^7 + x^6 + x^4 + x^3 + x^6 + x^5 + x^3 + x^2 =$$

$$x^{11} + (1+1)x^{10} + x^9 + x^8 + (1+1+1)x^7 + (1+1+1)x^6 + x^5 + x^4 + (1+1)x^3 + x^2 \pmod{2} =$$

$$x^{11} + x^9 + x^8 + x^7 + x^6 + x^5 + x^4 + x^2 / x^8 + x^4 + x^3 + x + 1 = x^3 + x + 1$$

$$+ x^{11} + x^9 + x^8 + x^7 + x^6 + x^5 + x^4 + x^2$$

1, everything cancels so checks out.

$$x^8 \equiv x^4 + x^3 + x + 1$$

$$A(x) * B(x)^{-1} \pmod{P(x)}$$

$$(x^4 + x + 1)(x^4 + x^3 + x + 1) = x^8 + x^7 + (1+1)x^5 + (1+1+1)x^4 + x^3 + (1+1)x + 1 \pmod{2} =$$

$$x^8 + x^7 + x^4 + x^3 + x^2 + 1 = (x^4 + x^3 + x + 1) + x^7 + x^4 + x^3 + x^2 + 1 = x^7 + (1+1)x^4 + (1+1)x^3 + x^2 + x + (1+1) =$$

$$x^7 + x^2 + x = 86$$

5. Consider the field GF (2⁴), with P(x) = x⁴+x+1 being the irreducible polynomial. Find the inverses of A(x) = x and B(x) = x² + x. You can find the inverses by trial and error, i.e., brute-force search (or by applying the Extended Euclidean Algorithm). Verify your answer by multiplying the inverses you determined by A and B, respectively.

$$P(x) = x^4 + x + 1$$

$$x^4 \equiv x + 1$$

$$A(x) = x$$

$$\text{Try: } A^{-1}(x) = x^3 + 1$$

$$x(x^3 + 1) = x^4 + x \equiv (x + 1) + x = 1$$

$$\text{So, } x^{-1} = x^3 + 1$$

$$A(x)^{-1} = x^3 + 1$$

$$B(x) = x^2 + x$$

$$\text{Try: } B^{-1}(x) = x^2 + x + 1$$

$$(x^2 + x)(x^2 + x + 1) = x^4 + (1+1)x^3 + (1+1)x^2 + x = x^4 + x \equiv (x + 1) + x = 1$$

$$\text{So, } x^{-1} = x^2 + x + 1$$

$$B(x)^{-1} = x^2 + x + 1$$

6. Find all irreducible polynomials of degree 3 over GF (2).

general polynomial of degree 3 over GF (2)

$$x^3 + ax^2 + bx + c, \dots, a, b, c, \in \{0,1\}$$

List all Possible polynomials



Evaluate at x=0, x = 1

$x^3 + 0x^2 + 0x + 0 = x^3$	$P(x) = x^3 : P(0) = 0$; Reducible
$x^3 + 0x^2 + 0x + 1 = x^3 + 1$	$P(x) = x^3 + 1 : P(1) = 1^3 + 1 = 2 \equiv 0 \pmod{2}$; Reducible
$x^3 + 0x^2 + 1x + 0 = x^3 + x$	$P(x) = x^3 + x : P(0) = 0 + 0 = 0$; Reducible
$x^3 + 0x^2 + 1x + 1 = x^3 + x + 1$	$P(x) = x^3 + x + 1 : P(1) = 1^3 + 1 + 1 = 3 \equiv 1 \pmod{2}$; Irreducible
$x^3 + 1x^2 + 0x + 0 = x^3 + x^2$	$P(x) = x^3 + x^2 : P(0) = 0 + 0$; Reducible
$x^3 + 1x^2 + 0x + 1 = x^3 + x^2 + 1$	$P(x) = x^3 + x^2 + 1 : P(1) = 1^3 + 1^2 + 1 = 3 \equiv 1 \pmod{2}$; Irreducible
$x^3 + 1x^2 + 1x + 0 = x^3 + x^2 + x$	$P(x) = x^3 + x^2 + x : P(0) = 0 + 0 + 0$; Reducible
$x^3 + 1x^2 + 1x + 1 = x^3 + x^2 + x + 1$	$P(x) = x^3 + x^2 + x + 1 : P(1) = 1^3 + 1^2 + 1 + 1 = 4 \equiv 0 \pmod{2}$; Reducible

So, all irreducible polynomials of degree 3 over GF (2) are: $x^3 + x + 1$ and $x^3 + x^2 + 1$.

7. We consider AES with 128-bit block length and 128-bit key length. What is the output of the first round of AES if the plaintext consists of 128 ones, and the first subkey also consists of 128 ones? You need to show each intermediate step!

11 = FF

$S(A_i(x)) = B_i(x)$; all 16 s-boxes are the same

$$P(x) = x^8 + x^4 + x^3 + x + 1$$

$$x^8 = x^4 + x^3 + x + 1$$

First Step: Add round key

Plaintext $\oplus$ Round key

Plaintext = 128 1's = FF

First round key = 128 1's = FF

$FF \oplus FF = 00$; all sums result in 00

Second Step: Sub Bytes

Since all bytes are 00; Use S-Box Table for value (0,0) = 63 hex

All bytes are now replaced with 63 hex.

Third Step: Shift Rows

63	63	63	63
63	63	63	63
63	63	63	63
63	63	63	63

→

63	63	63	63
63	63	63	63
63	63	63	63
63	63	63	63

First row: no shift

Second row: left shift 1

Third row: left shift 2

Fourth row: left shift 3

Since all bytes are identical, row rotations don't change the state. All bytes are 63.

Fourth Step: Mix Columns

M =

02	03	01	0	63
01	02	03	01	63
01	01	02	03	63
03	01	01	02	63

Row 1: $(02 * 63) \oplus (03 * 63) \oplus (01 * 63) \oplus (01 * 63)$

$(C6) \oplus (A5) \oplus (63) \oplus (63)$

$C6 \oplus A5 = 63$

Same occurs for row 2,3,4; all result in 63. So, because all four columns are identical, no change happens.

Fifth Step: Add Round Key

Round 1 key = FF

All Bytes Current State = 63

$63 \oplus FF = 9C$

The output from the first round of AES is 16 bytes of 9C or (10011100) repeating.

8. We consider the first part of the ByteSub operation, i.e, the Galois field inversion.

8.1. Using Table 4.2 in the text book, what is the inverse of the bytes 29, F3 and 01, where each byte is given in hexadecimal notation?

29 inverse = 0A

F3 inverse = 34

01 inverse = 01

8.2. Verify your answer by performing a $GF(2^8)$ multiplication with your answer and the input byte. Note that you have to represent each byte first as polynomials in $GF(2^8)$. The MSB of each byte represents the x^7 coefficient.

$29 = (0010\ 1001) \rightarrow A(x) = x^5 + x^3 + 1$

$0A = (0000\ 1010) \rightarrow B(x) = x^3 + x$

$(x^5 + x^3 + 1)(x^3 + x) = (x^8 + x^6 + x^3) + (x^6 + x^4 + x) = x^8 + x^4 + x^3 + x$

$P(x) = x^8 + x^4 + x^3 + x + 1$

$x^8 = x^4 + x^3 + x + 1$

$$(x^4 + x^3 + x + 1) + x^4 + x^3 + x = (1+1)x^4 = (1+1)x^3 + (1+1)x + 1 = 1$$

$$\text{So } 29 * 0A \equiv 1$$

$$F3 = (1111\ 0011) = x^7 + x^6 + x^5 + x^4 + x + 1$$

$$34 = (0011\ 0100) = x^5 + x^4 + x^2$$

$$(x^7 + x^6 + x^5 + x^4 + x + 1) (x^5 + x^4 + x^2) = (x^{12} + x^{11} + x^9) + (x^{11} + x^{10} + x^8) + (x^{10} + x^9 + x^7) + (x^9 + x^8 + x^6) + (x^6 + x^5 + x^3) + (x^5 + x^4 + x^2) =$$

$$x^{12} + x^9 + x^7 + x^4 + x^3 + x^2$$

$$x^8 \equiv x^4 + x^3 + x + 1$$

$$x^9 \equiv x^5 + x^4 + x^2 + x$$

$$x^{12} \equiv x^8 + x^7 + x^5 + x^3$$

$$(x^8 + x^7 + x^5 + x^3) + (x^5 + x^4 + x^2 + x) + x^7 + x^4 + x^3 + x^2 = x^8 + x^4 + x^3 + x =$$

$$(x^4 + x^3 + x + 1) + x^4 + x^3 + x = 1$$

$$\text{So } F3 * 34 \equiv 1$$

$$01 = (0000\ 0001) = 1$$

$$01 = (0000\ 0001) = 1$$

$$1 * 1 = 1, \text{ same value.}$$

$$\text{So, } 01 * 01 \equiv 1$$

9. Your task is to compute the S-Box, i.e., the ByteSub, values for the input bytes 19, F4 and 02, where each byte is given in hexadecimal notation.

9.1. First, look up the inverses using Table 4.2 in the text book to obtain values B. What are the inverted bytes?

Inverse of 19: 3F

Inverse of F4: 68

Inverse of 02: 8D

9.2. Now, perform the affine mapping by computing the matrix–vector multiplication and addition. What are the three resulting bytes? Make sure to verify your result using the S-Box Table 4.3.3 in the text book.

$y = (y_7, \dots, y_0)$ be the inverse byte, S-box output $s = (s_7, \dots, s_0)$ is $s = A y \oplus c$,

$$A = \begin{array}{|c|c|c|c|c|c|c|c|} \hline 1 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ \hline 1 & 1 & 0 & 0 & 0 & 1 & 1 & 1 \\ \hline 1 & 1 & 1 & 0 & 0 & 0 & 1 & 1 \\ \hline 1 & 1 & 1 & 1 & 0 & 0 & 0 & 1 \\ \hline 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ \hline 0 & 1 & 1 & 1 & 1 & 1 & 0 & 0 \\ \hline 0 & 0 & 1 & 1 & 1 & 1 & 1 & 0 \\ \hline 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 \\ \hline \end{array} \quad C = \begin{array}{|c|} \hline 0 \\ \hline 1 \\ \hline 1 \\ \hline 0 \\ \hline 0 \\ \hline 0 \\ \hline 1 \\ \hline 1 \\ \hline \end{array}$$

For each bit

$$S_0 = y_0 \oplus y_4 \oplus y_5 \oplus y_6 \oplus y_7 \oplus c_0$$

$$S_1 = y_1 \oplus y_5 \oplus y_6 \oplus y_7 \oplus y_0 \oplus c_1$$

$$S_2 = y_2 \oplus y_6 \oplus y_7 \oplus y_0 \oplus y_1 \oplus c_2$$

$$S_3 = y_3 \oplus y_7 \oplus y_0 \oplus y_1 \oplus y_2 \oplus c_3$$

$$S_4 = y_4 \oplus y_0 \oplus y_1 \oplus y_2 \oplus y_3 \oplus c_4$$

$$S_5 = y_5 \oplus y_1 \oplus y_2 \oplus y_3 \oplus y_4 \oplus c_5$$

$$S_6 = y_6 \oplus y_2 \oplus y_3 \oplus y_4 \oplus y_5 \oplus c_6$$

$$S_7 = y_7 \oplus y_3 \oplus y_4 \oplus y_5 \oplus y_6 \oplus c_7$$

19

Inverse: 3F = (0011 1111)

$$y_7 \dots y_0 = 0, 0, 1, 1, 1, 1, 1, 1$$

$$s_0 = 0, s_1 = 0, s_2 = 1, s_3 = 0, s_4 = 1, s_5 = 0, s_6 = 1, s_7 = 1$$

$$s = 1101 \ 0100 = \mathbf{D4}$$

F4

Inverse: 68 = (0110 1000)

$$y_7 \dots y_0 = 0, 1, 1, 0, 1, 0, 0, 0$$

$$s_0 = 1, s_1 = 1, s_2 = 1, s_3 = 1, s_4 = 1, s_5 = 1, s_6 = 0, s_7 = 1$$

$$s = 1011 \ 1111 = \mathbf{BF}$$

02

Inverse: 8D = (1000 1101)

$$y_7 \dots y_0 = 1, 0, 0, 0, 1, 1, 0, 1$$

$$s_0 = 1, s_1 = 1, s_2 = 1, s_3 = 0, s_4 = 1, s_5 = 1, s_6 = 1, s_7 = 0$$

$$s = 0111\ 0111 = 77$$

10. We now look into the diffusion properties of AES after a single round. Let $W = (w_0, w_1, w_2, w_3) = (0x01000000, 0x00000000, 0x00000000, 0x00000000)$ be the input in 32-bit chunks to a 128-bit AES. The subkeys for the computation of the result of the first round of AES are $W_0, \dots, W_7$ with 32 bits each are given by:

$$W_0 = (0x2B7E1516),$$

$$W_1 = (0x28AED2A6),$$

$$W_2 = (0xABF71588),$$

$$W_3 = (0x09CF4F3C),$$

$$W_4 = (0xA0FAFE17),$$

$$W_5 = (0x88542CB1),$$

$$W_6 = (0x23A33939),$$

$$W_7 = (0x2A6C7605).$$

10.1. Compute the output of the first round of AES to the input W and the subkeys $W_0, \dots, W_7$.

$$s[r,c] \text{ rows } = 0..3, \text{ columns } c = 0..3.$$

Step 1 Add Round key

Input W in bytes: only $b_0 = 01$, everything else 00

01	00	00	00
00	00	00	00
00	00	00	00
00	00	00	00

$$W_0 = 2B, W_1 = 28, W_2 = AB, W_3 = 09$$

Round key =

2B	28	AB	09
7E	AE	F7	CF
15	D2	15	4F
16	A6	88	3C

2A	28	AB	09
7E	AE	F7	CF
15	D2	15	4F

16	A6	88	3C
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State $\oplus$ Round key

$S_0 =$

Step 2 Sub Bytes

SubBytes (S_0) =

E5	34	62	01
F3	E4	68	8A
59	B5	59	84
47	24	C4	EB

Step 3 Shift rows

Row 0: no shift

Row 1: shift left 1

Row 2: shift left 2

Row 3: shift left 3

ShiftRows =

E5	E4	59	EB
34	68	84	47
62	8A	59	24
01	F3	B5	C4

Step 4 Mix Columns

For each column $[a_0, a_1, a_2, a_3]^T$, compute

$$b_0 = 2a_0 \oplus 3a_1 \oplus a_2 \oplus a_3$$

$$b_1 = a_0 \oplus 2a_1 \oplus 3a_2 \oplus a_3$$

$$b_2 = a_0 \oplus a_1 \oplus 2a_2 \oplus 3a_3$$

$$b_3 = 3a_0 \oplus a_1 \oplus a_2 \oplus 2a_3$$

MixColumns =

54	13	3C	7D
36	34	A2	FC
95	86	36	D4
44	3E	3D	D6

Step 5 Add Round key

Round Key 1 =

A0	88	23	2A
FA	54	A3	6C
FE	2C	39	76
17	B1	39	05

Final XOR:

State after Round 1 =

F4	9B	1F	57
CC	60	01	90
6B	AA	0F	A2
53	8F	04	D3

Output: F4 CC 6B 53 9B 60 AA 8F 1F 01 0F 57 90 A2 D3

10.2. Compute the output of the first round of AES for the case that all input bits are zero.

Step 1 Add Round key

Input = all zeroes so $\oplus$ just copies the key bytes into the states

Round key =

2B	28	AB	09
7E	AE	F7	CF
15	D2	15	4F
16	A6	88	3C

State $\oplus$ Round key

S₀ =

2A	28	AB	09
7E	AE	F7	CF
15	D2	15	4F
16	A6	88	3C

Step 2 Sub Bytes

SubBytes (S₀) =

F1	34	62	01
F3	E4	68	8A
59	B5	59	84
47	24	C4	EB

Step 3 Shift rows

Row 0: no shift

Row 1: shift left 1

Row 2: shift left 2

Row 3: shift left 3

ShiftRows =

F1	E4	59	EB
34	68	84	47
62	8A	59	24
01	F3	B5	C4

Step 4 Mix Columns

For each column $[a_0, a_1, a_2, a_3]^T$, compute

$$b_0 = 2a_0 \oplus 3a_1 \oplus a_2 \oplus a_3$$

$$b_1 = a_0 \oplus 2a_1 \oplus 3a_2 \oplus a_3$$

$$b_2 = a_0 \oplus a_1 \oplus 2a_2 \oplus 3a_3$$

$$b_3 = 3a_0 \oplus a_1 \oplus a_2 \oplus 2a_3$$

MixColumns =

7C	22	81	78
13	34	86	3E
3C	A2	36	3D
7D	FC	D4	D6

Step 5 Add Round key

Round Key 1 =

A0	88	23	2A
FA	54	A3	6C
FE	2C	39	76
17	B1	39	05

Final XOR:

State after Round 1 =

DC	9B	1F	57
D8	60	01	90
7F	AA	0F	A2
6F	8F	04	D3

Output: DC D8 7F 6F 9B 60 AA 8F 1F 01 0F 04 57 90 A2 D3

10.3. How many output bits have changed? Note that we only consider a single round - after every further round, more output bits will be affected ("avalanche effect").

There was 10 bits that changed.

11. The MixColumn transformation of AES consists of a matrix–vector multiplication in the field $GF(2^8)$ with $P(x) = x^8 + x^4 + x^3 + x + 1$. Let $b = (b_7x^7 + \dots + b_0)$ be one of the (four) input bytes to the vector–matrix multiplication. Each input byte is multiplied with the constants 01, 02 and 03. Your task is to provide exact equations for computing those three constant multiplications. We denote the result by $d = (d_7x^7 + \dots + d_0)$.

11.1. Equations for computing the 8 bits of $d = 01 \cdot b$.

Since the byte is unchanged, Each output bit equals the input bit.

$$d_i = b_i \text{ for } i = 0, \dots, 7$$

11.2. Equations for computing the 8 bits of $d = 02 \cdot b$.

$02 * b(x) = x * b(x)$; left-bit shift of byte

$$d = (b_6, b_5, b_4, b_3, b_2, b_1, b_0, 0)$$

$$\text{let } b = (b_7, b_6, b_5, b_4, b_3, b_2, b_1, b_0)$$

$$\text{then } d = (d_7, d_6, d_5, d_4, d_3, d_2, d_1, d_0)$$

$$d_7 = b_6$$

$$d_6 = b_5$$

$$d_5 = b_4 \oplus b_7$$

$$d_4 = b_3 \oplus b_7$$

$$d_3 = b_2$$

$$d_2 = b_1$$

$$d_1 = b_0 \oplus b_7$$

$$d_0 = b_7$$

11.3. Equations for computing the 8 bits of $d = 03 \cdot b$.

$$03 * b = (02 * b) \oplus (01 * b):$$

Let $02 * b$ bits be as above, then

$$d_7 = b_6 \oplus b_7$$

$$d_6 = b_5 \oplus b_6$$

$$d_5 = b_4 \oplus b_5$$

$$d_4 = (b_3 \oplus b_7) \oplus b_4$$

$$d_3 = (b_2 \oplus b_7) \oplus b_3$$

$$d_2 = b_1 \oplus b_2$$

$$d_1 = (b_0 \oplus b_7) \oplus b_1$$

$$d_0 = b_7 \oplus b_0$$

12. Derive the bit representation for the following round constants within the key schedule:

12.1. RC[8]

RC[8]: $r_8 = 80$

Byte bits: 10000000

12.2. RC[9]

RC[9]: $r_9 =$

$$x * x^7 = x^8 \equiv x^4 + x^3 + x + 1$$

$$\text{So, } r_9 = x^4 + x^3 + x + 1$$

Left shift 80 $\rightarrow$ 00 (MSB = 1)

$$\text{Reduce } 00 \oplus 1B = 1B$$

$$\text{So, } r_9 = 1B$$

Byte Bits: 00011011

12.3. RC[10]

RC[10]: $r_{10} =$

$$x(x^4 + x^3 + x + 1) = x^5 + x^4 + x^2 + x$$

Degree is $5 < 8$, so no reduction needed.

$$\text{So, } r_{10} = x^5 + x^4 + x^2 + x$$

Left shift 1B $\rightarrow$ 36 (MSB = 0)

$$\text{So, } r_{10} = 36$$

Byte Bits: 00110110